Measures of Association: Nominal Variables

I. Clinical Trials vs. Epidemiological Studies

II. Relative Risk

III. NNT & NNH

IV. Odds Ratio

V. More Examples

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	z Test Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5) Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.
†Not covered in this text.
‡Or interval data that are not necessarily normally distributed.

I. Clinical Trials vs. Epidemiological Studies

A. Clinical Trials

Subjects are randomized to treatment groups and then the outcome is observed following assignment (prospective)

B. Epidemiological Studies

Subjects not randomized to treatment groups.

1) Prospective (cohort) – Enrolled in study then followed over time to determine outcome

2) Retrospective (case-control) – People who have the outcome of interest are identified and then grouped depending on whether they were exposed to the risk factor or not

Association

- A relationship between 2 measured quantities renders them statistically dependent
- The term “association” is used to distinguish a relationship from being a causal one (i.e., if two quantities are associated it does not imply that either one causes the other)
- If an association is linear, the term “correlation” is used
- How is the degree of association quantified when the variables are nominal?

II. Relative Risk or Risk Ratio (RR)

The degree of association between nominal variables can be quantified by calculating the Relative Risk if the study is:

1) A Clinical Trail or
2) A Prospective Epidemiological Study

TABLE 5-13. Arrangement of Data to Compute Relative Risk

Sample Group	Number of People		
	Disease	No Disease	Total
Treated (or exposed to risk factor)	n_{TD}	n_{TN}	n_T
Control (or not exposed to risk factor)	n_{CD}	n_{CN}	n_C
Total	n_D	n_N	

$$RR = \frac{Probability\ of\ event\ in\ treatment\ group}{Probability\ of\ event\ in\ control\ group} = \frac{n_{TD}/n_T}{n_{CD}/n_C}$$

Table 5-1 Thrombus Formation in People Receiving Dialysis and Treated with Placebo or Aspirin

	Number of patients		
Sample group	Developed thrombi	Free of thrombi	Treated
Placebo	18 n_{CD}	7 n_{CN}	25 n_C
Aspirin	6 n_{TD}	13 n_{TN}	19 n_T
Total	24	20	44

Source: H. R. Harter, J. W. Burch, P. W. Majerus, N. Stanford, J. A. Delmez, C. B. Anderson, and C. A. Weerts "Prevention of Thrombosis in Patients on Hemodialysis by Low-Dose Aspirin," *N. Engl. J. Med.*, **301**:577–579, 1979. Reprinted by permission of the *New England Journal of Medicine.*

Event: Thrombi

$$\hat{p}_{asp} = n_{TD}/n_T = 6/19 = 0.316 \quad \hat{p}_{pla} = n_{CD}/n_C = 18/25 = 0.720$$

$$RR = \frac{\hat{p}_{asp}}{\hat{p}_{pla}} = \frac{0.316}{0.720} = 0.439$$

Absolute risk reduction: $\hat{p}_{pla} - \hat{p}_{asp} = 0.720 - 0.316 = 0.404$

$$RR = \frac{\hat{p}_{asp}}{\hat{p}_{pla}} = \frac{0.316}{0.720} = 0.439$$

The relative risk of developing thrombi in the aspirin group is approximately 0.44 times the risk of developing thrombi in the placebo group

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Absolute risk reduction: $\hat{p}_{pla} - \hat{p}_{asp} = 0.720 - 0.316 = 0.404$

There is an approximately 40% absolute risk reduction for developing thrombi in the aspirin group versus the placebo group

III. Number Needed to Treat (NNT)

- Number of patients that need to be treated in order for one patient to benefit
- Used when treatment increases likelihood of good outcome or decreases likelihood of bad outcome
- Calculated as the inverse of the absolute difference in prevalence of the outcome in the two groups:

$$NNT = \frac{1}{\left|\hat{p}_1 - \hat{p}_2\right|}$$

- Lower NNT means treatment is more effective; higher NNT means treatment is less effective

III. Number Needed to Treat (NNT)

- From placebo/aspirin example:

$$NNT = \frac{1}{\left|\hat{p}_{pla} - \hat{p}_{asp}\right|} = \frac{1}{\left|0.720 - 0.316\right|} = \frac{1}{0.404} = 2.48$$

Approximately 2.5 patients need to be treated with aspirin to have one additional patient avoid thrombosis

III. Number Needed to Harm (NNH)

- Number of patients that need to be treated/exposed in order for <u>one patient</u> to be harmed
- Used when exposure increases likelihood of bad/dangerous outcome or decreases likelihood of good outcome
- Calculated as the inverse of the absolute difference in prevalence of the outcome in the two groups:

$$NNH = \frac{1}{\left|\hat{p}_1 - \hat{p}_2\right|}$$

- Lower NNH means treatment is more harmful; higher NNH means treatment is less harmful

III.

Idea of NNT and NNH

- Suppose a drug reduces your chances of dying from Hogwart's disease by 20 percent.
- Normally, 10 out of every 1,000 people who get Hogwart's disease die from it.
- If every single infected patient goes on the drug, now only eight in 1,000 will die from Hogwart's disease (thereby saving 2 people per 1000 infected patients).
- So, for every 500 patients who get the drug, one will be spared death by Hogwart's disease, meaning NNT is 500.
- But what if the drug's NNH is 20 and the unwanted side effect is severe?
- In that case, $500/20 = 25$ patients will suffer serious harm for each one who is saved.

From: https://www.propublica.org/article/when-evidence-says-no-but-doctors-say-yes

Retrospective Study:

- Done after the fact (after the disease has developed)
- People who already have outcome of interest (e.g., disease) are identified and grouped based on whether or not they were exposed to the risk factor or treatment
- These people are paired with "case controls" to produce a suitable control group of people without the disease that come from the same population(s) as the cases; they are matched based on age, sex, ethnicity, socioeconomic status, etc.
- Some controls have been exposed and some have not been exposed to the risk factor or treatment: selection of controls should be **INDEPENDENT** of exposure status
- Studies often include more than one control per case to increase the sample size
- We do not know the exposure distribution for the entire source population so we cannot compute the probability of disease in each exposure-status group, but we can compute the odds of exposure in each outcome group
- The nature of retrospective data does not give us a sense of the "true" population rates, which is why we can only calculate the odds ratio (and not relative risk) from this type of data

Odds Ratio (OR)

The degree of association between nominal variables can be quantified by calculating the Odds Ratio if the study is:

- A Retrospective Epidemiological Study
  (can also be used for either Prospective Study design)

TABLE 5-14. Arrangement of Data to Compute Odds Ratio

Sample Group	Number of People: Disease "Cases"	Number of People: No Disease "Controls"
Exposed to risk factor (or treatment)	n_{ED}	n_{EN}
Unexposed to risk factor (or treatment)	n_{UD}	n_{UN}
Total	n_D	n_N

Same calculation

$$OR = \frac{Odds\ of\ exposure\ in\ cases}{Odds\ of\ exposure\ in\ controls} = \frac{n_{ED}/n_{UD}}{n_{EN}/n_{UN}} = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}}$$

$$OR = \frac{Odds\ of\ cases\ in\ those\ exposed}{Odds\ of\ cases\ in\ those\ not\ exposed} = \frac{n_{ED}/n_{EN}}{n_{UD}/n_{UN}} = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}}$$

(prospective data only)

Table 5-16 Passive Smoking and Breast Cancer

Sample group	Number of People: Cases (breast cancer)	Number of People: Controls
Exposed to secondhand smoke	50 n_{ED}	43 n_{EN}
Not exposed to secondhand smoke	14 n_{UD}	35 n_{UN}
Total	64	78

First: z or χ^2 test $\Rightarrow$ reject H_0 with $P<0.01$

$$OR = \frac{Odds\ of\ secondhand\ smoke\ exposure\ in\ women\ with\ breast\ cancer}{Odds\ of\ secondhand\ smoke\ exposure\ in\ controls}$$

$$OR = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}} = \frac{50 \cdot 35}{14 \cdot 43} = 2.91$$

The odds of exposure to secondhand smoke are 2.91 times higher in patients with breast cancer than the odds of exposure in disease-free subjects

Oral Contraceptive and Myocardial Infarction

(From: B. Rosner, *Fund of Biostats*)

MI status over 3 years

OC-use group	Yes	No	Total
OC users	13	4987	5,000
Non-OC users	7	9993	10,000
Total	20	14,980	15,000

- OC: oral contraceptive
- MI: myocardial infarction

First: z or χ^2 test $\Rightarrow$ reject H_0 with $P<0.005$

What is the odds ratio associating the development of an MI with being an OC user versus not being an OC user?

$$OR = \frac{Odds\ of\ exposure\ in\ cases}{Odds\ of\ exposure\ in\ controls} = \frac{n_{ED}/n_{UD}}{n_{EN}/n_{UN}} = \frac{n_{ED}n_{UN}}{n_{UD}n_{EN}} = \frac{n_{ED}/n_{EN}}{n_{UD}/n_{UN}}$$

$$OR_{OCvsNOC} = \frac{n_{ED} \cdot n_{UN}}{n_{UD} \cdot n_{EN}} = \frac{(13)(9993)}{(7)(4987)} = 3.72$$

So, the odds of having taken an OC are 3.72 times greater for those who suffered an MI versus those who did not suffer an MI

OR

The odds of developing an MI are 3.72 times greater for an OC user versus a non-OC user

V. More Examples

5-13 Many postmenopausal women are faced with the decision of whether they want to take hormone replacement therapy or not. Benefits of hormone replacement include decreased risk of cardiovascular disease and osteoporosis. However, hormone replacement therapy has also been associated with increased risk of breast cancer and endometrial cancer. Francine Grodstein and colleagues* investigated the relationship between hormone replacement therapy and overall mortality in a large group of postmenopausal women. The women used in this study were selected from a sample of registered nurses participating in the Nurses' Health Study. This study has been tracking the health status of a large group of registered nurses since 1976, updating information every 2 years. Women became eligible for Grodstein's study when they became menopausal and were included as long as they did not report a history of cardiovascular disease or cancer on the original 1976 questionnaire. Based on the data in Table 5-25, is there any evidence that the risk of death differs in women who were identified as currently using hormone replacement therapy?

TABLE 5-25. Current Use of Hormone Replacement Therapy and Survival

	Number of People	
	Deceased	Alive
Currently using hormone replacement therapy	574	8483
Never used hormone replacement therapy	2051	17,520

If there is an association, quantify the degree of association between dying and taking hormone replacement therapy

Session 13 Lecture Exercise

5-11 Cigarette smoking is associated with increased incidence of many types of cancers. Jian-Min Yuan and colleagues[‡] wanted to investigate whether cigarette smoking was also associated with increased risk of renal cell cancer. They recruited patients with renal cell cancer from the Los Angeles County Cancer Surveillance Program to serve as cases Control subjects without renal cell cancer were matched on sex, age (within 5 years), race, and neighborhood of residence to each case subject. After recruiting a total of 2314 subjects for the study, Yuan and colleagues visited subjects in their homes and interviewed them about their smoking habits, both past and present (Table 5-23). What effect does smoking cigarettes have on the risk of developing renal cell cancer?

If there is an association, quantify the degree of association between developing renal cell cancer and ever smoking

TABLE 5-23. Smoking and Renal Cell Cancer

	Number of People	
	Renal Cell Cancer	No Cancer
Ever smoked cigarettes	800	713
Never smoked cigarettes	357	444

χ^2 Test in R

Generic function

- ***chisq.test()***: performs chi-squared calculation on given contingency table
  - Can use *table()* or *xtabs()* (or similar) to create contingency tables

From the "lsr" package

- ***goodnessOfFitTest()***: compares given/observed frequency-vector against null-hypothesis of equal probabilities for each category OR a user-specified probability vector